BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Species Used *Staphylococcus aureus,* RN4220

Molecules Electroporated DNA: pC194 (2.9 kB), pE194 (2.9 kB), pI258 (2.9 kB), pMH109 (7.4 kB), pMH120 (7.8 kB), supercoiled.

Before the Pulse

Cell Growth Medium B2 broth (see notes)

Growth Phase at Harvest Mid-log; 260 Klett units, #66 red filter.

Pre-pulse Incubation None

Wash Solution See notes

The Pulse

Instruments Used Gene Pulser® apparatus & Pulse Controller

Electroporation Temperature Room temperature, 20 °C

Electroporation Medium 10% glycerol

Cuvette Gap 0.1 cm

Voltage 2.3 kV

Cell Density 3 x 10 (10) cells / ml

Volume of Cells 60 µl

Field Strength 23 kV/cm

DNA Concentration 17 µg DNA / µl

Capacitor 25 µF

DNA Resuspension Buffer B2 broth

Resistor (Pulse Controller) 100 Ω

Volume of DNA 1 to 6 µl

Time Constant 2.5 msec

After the Pulse

Outgrowth Medium NYE broth: 1% casein hydrolyzate (Sigma), 0.5% yeast extract, 0.5% NaCl

Outgrowth Temperature 37 ° C

Length of Incubation 2 hours

Selection Method or Assay Used NYE agar and appropriate selective agent

Electroporation Efficiency 2 x 10 (8) CFU per µg DNA

Per Cent Survival 2 %

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
B2 broth = 1% casein hydrolyzate (Sigma), 2.5% yeast extract (Difco), 0.5% glucose, 0.1% KHPO4, 0.5% NaCl pH 7.5, overnight culture diluted 1/25 in 25 ml B2 broth in 300 ml Klett flask.
Wash Solution: 3 washes deionized water; 1/5 wash 10% glycerol; 1/10 wash 10% glycerol (Wash volume refers to volume of growth medium).

Name of Submittor Richard A. Laddaga, Ph.D. Asst. Prof.

Institution Address Bowling Green State University
Biological Sciences
Bowling Green, OH 43403-0212

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